

NeurOncoVision: Building A Novel Multi-Modal Machine Learning Framework for Brain Tumor Detection, Classification, and Prediction

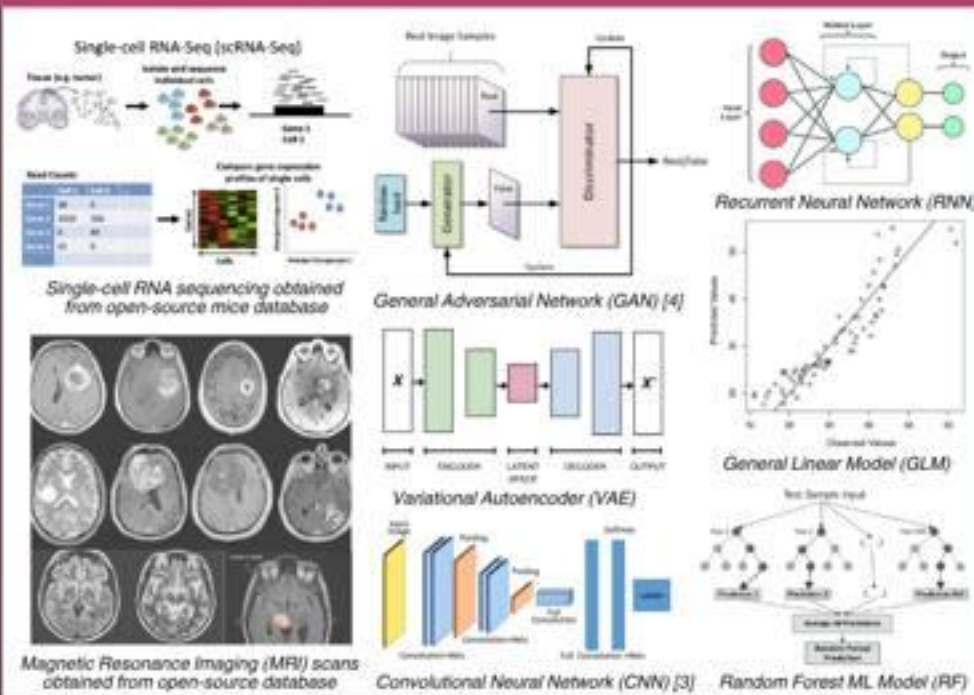
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OVERVIEW

OBJECTIVE: Integrate synthetic data generation and deep learning models for comprehensive brain tumor analysis

- Identify **key genetic drivers** and link **developmental trajectories** to clinical outcomes
- Use machine learning to enable further machine learning analysis on brain tumors

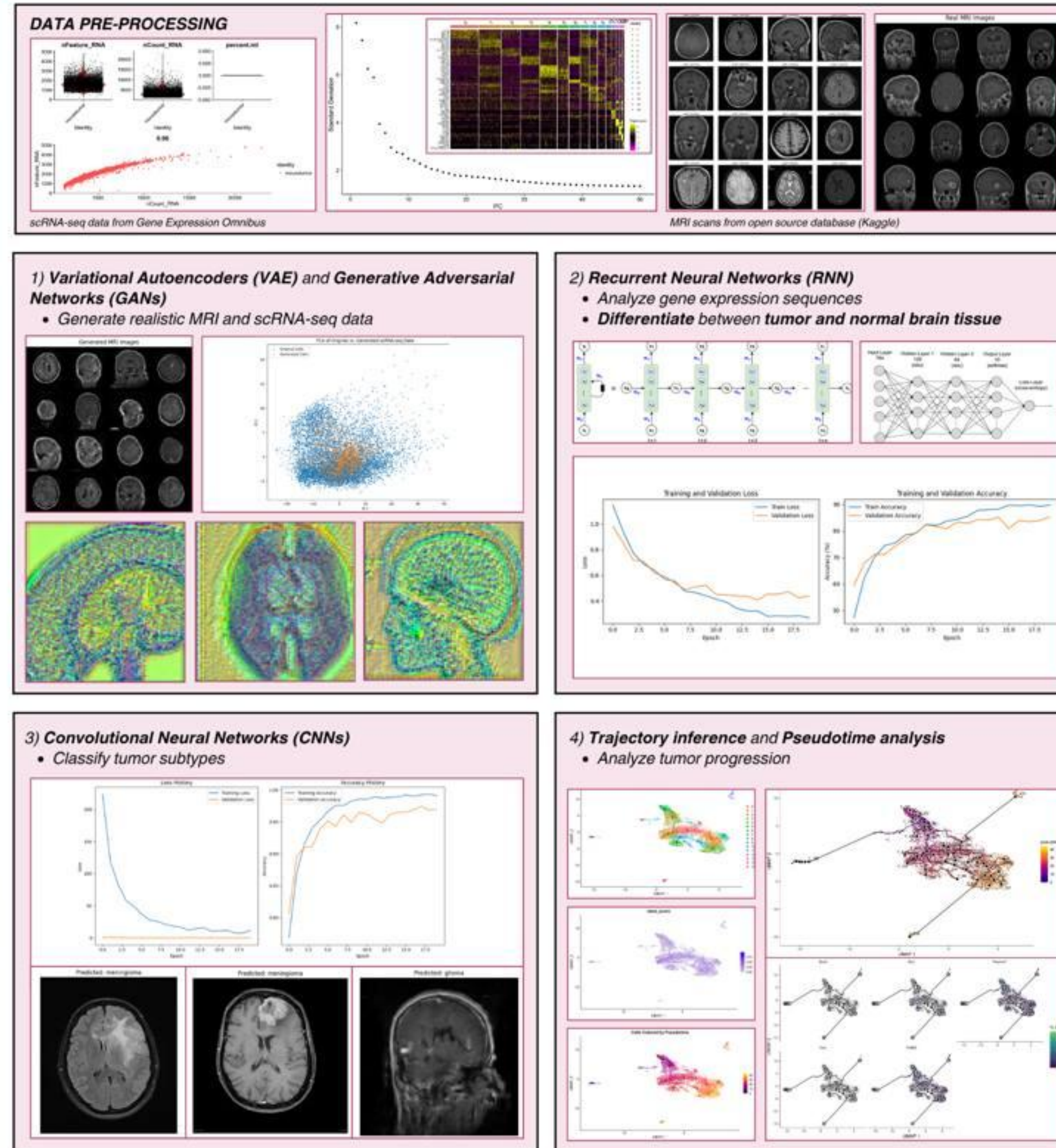
BACKGROUND



PROJECT METHODOLOGY

- 5-PRONGED APPROACH**
- Employ **Variational Autoencoders (VAE)** and **Generative Adversarial Networks (GANs)** to generate realistic MRI and scRNA-seq data.
 - Enhance **detection** by using **Recurrent Neural Networks (RNN)** to analyze gene expression sequences and **differentiate between tumor and normal brain tissue**.
 - Improve **classification** with **Convolutional Neural Networks (CNNs)** trained on both real and synthetic MRI data to **distinguish tumor subtypes**.
 - Use **trajectory inference** and **pseudotime analysis** to analyze tumor progression and identify high-risk developmental pathways and potential therapeutic targets.
 - Apply **traditional machine learning methods**, General Linear Models (GLMs) and Random Forest (RF), for tumor classification using **real and simulated scRNA-seq data**.

RESULTS



5) Traditional **Machine Learning** methods: **General Linear Models (GLMs)** and **Random Forest (RF)**

- Validating our approach



CONCLUSION

- Leveraged advanced neural networks (VAEs, GANs, RNNs, CNNs) to generate synthetic data and distinguish tumor characteristics
- Identified high-risk genetic pathways and demonstrated potential for improving diagnostic precision and personalized treatment strategies
- Successfully integrated machine learning and statistical methods to provide comprehensive tumor analysis

FUTURE WORK

- Enhancing synthetic data generation by improving resolution and fidelity of MRI and single-cell RNA sequencing data
- Expanding dataset to include more comprehensive human single-cell transcriptomics and clinical MRI scans
- Aim to validate machine learning models in more robust, real-world clinical settings
- Next phase focused on refining NeurOncoVision framework to increase diagnostic and predictive capabilities

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